

2021

(T30 Battery & Charger)
Repair Training

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THE FUTURE OF POSSIBLE

01 Product Introduction

01 Product Introduction — T30 & T10 Battery



T30 Intelligent Flight Battery

- Model: BAX501-29000mAh-51.8V
- Weight: Approx. 10.1 kg
- Discharging Rate: 11.5C
- IP Rating: **IP54 + Metal Shell & Rubber Cap Protection**
- Capacity: 29000 mAh
- Voltage: 51.8 V
- Connector: **10-pin gold plating**



T10 Intelligent Flight Battery

- Model: BAX501-9500mAh-51.8V
- Weight: Approx. 3.8 kg
- Discharging Rate: 11.5C
- IP Rating: IP54 + Metal Shell & Rubber Cap Protection
- Capacity: 9500 mAh
- Voltage: 51.8 V
- Connector: **8-pin gold plating**

01 Product Introduction — T30 & T10 Battery Station



T30 Intelligent Battery Station

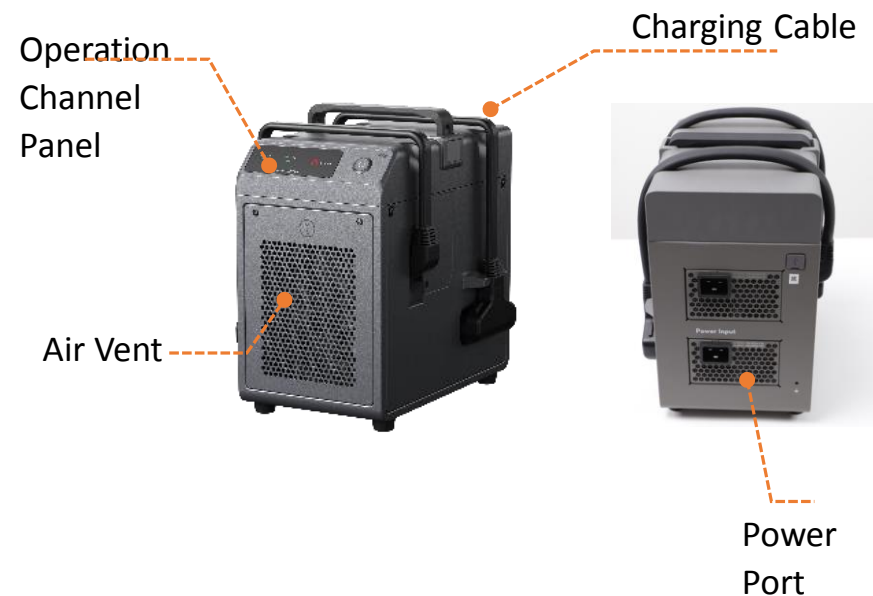
- Dimensions: 300×280×230 mm
- Optimized power module assembly for easy repair
- Potting process for main components to increase protection
- Power adaptive; improved charging safety
- Charging time: 9-12 minutes
- Two power modules



T10 Intelligent Battery Station

- Dimensions: 300×280×230 mm
- Optimized power module assembly for easy repair
- Potting process for main components to increase protection
- Power adaptive; improved charging safety
- Charging time: 7-10 minutes
- 1-piece power module

01 Product Introduction — Product Comparison – Charger



T30 Intelligent Battery Station

- Two charging cables
- Two power modules
- Quick-release power module
- Charging cable adapter cannot be quickly removed.



T10 Intelligent Battery Station

- Two charging cables
- 1-piece power module
- Quick-release power module
- Charging cable adapter: Not support quick-release



4-Channel Intelligent Battery Charger

- 4 charging cables
- 1-piece power module
- Power module cannot be quickly removed
- Quick-release charging cable adapter

01 Product Introduction — Specifications – Battery

Specifications	T30 Intelligent Flight Battery (BAX501-29000mAh-51.8V)	T10 Intelligent Flight Battery (BAX501-9500mAh-51.8V)
Nominal Voltage	51.8 V	
Battery Type	Lithium Battery 14S	
Discharge Rate	11.5 C	
Operating Temperature Range	-5° C to 45° C (23° F to 113° F)	
Charging Temperature Range	0° C to 60° C (32° F to 113° F)	
Warranty Details	1,000 charge cycles or 12 months (whichever is sooner)	
IP Rating	IP54 + Board-level potting protection	
Compatible Aircraft	DJI Agras T30	DJI Agras T10
Compatible Charging Device	T30 Intelligent Battery Station, D9000i Multifunction Inverter Generator	T10 Intelligent Battery Station, D4500i Multifunction Inverter Generator
Capacity	29000 mAh	9500 mAh
Energy	1502 Wh	430 Wh
Max Charging Power	7200 W	3600 W
Battery Charging Time*	9-12 mins when the battery station is connected with two AC power cables; 18-25 mins when the battery station is connected with one AC power cable	7-10 mins
Weight	Approx. 10.2 kg	Approx. 3.8 kg

01 Product Introduction — Specifications – Battery Station

Specifications	T30 Intelligent Battery Station (Model: CHX503-7000)	T10 Intelligent Battery Station (Model: CHX503-3500)
Operating Temperature Range	-5° to 45° C (23° to 113° F)	
Dimensions	300×280×230 mm	
Output Channel	2	
Compatible Battery	T30 Intelligent Flight Battery	T10/T16/T20 Intelligent Flight Battery
Weight	Approx. 5.5 kg (without CPX503-3000 Switching Power Supply)	Approx. 5.1 kg (without CPX503-3000 Switching Power Supply)

01 Product Introduction — In the Box



Battery



Battery Station



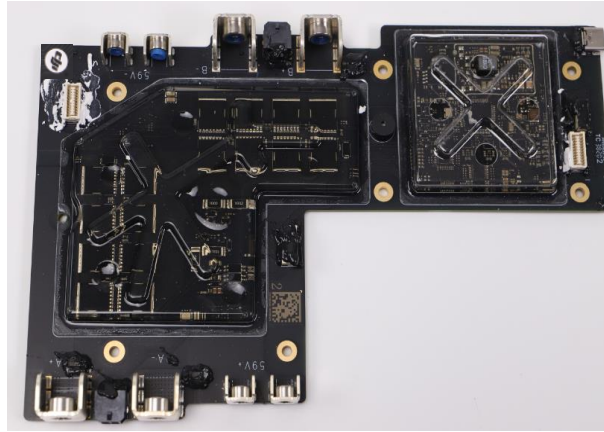
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02 Material Identification

02 Material Identification — Battery Station



LED Board



Battery Station Main Board



Power Button Module

LED board: Connects the management board and the system switch respectively, and transmits the switch signal to the management board through the LED board. The LED on the LED board is used to display the battery charging status and the status of the battery station.

Battery Station Main Board: Communicates with the power module and battery, manages the battery charging current, and outputs the LED signal to control the LED display. The management board firmware can be updated by connecting to the computer via Type-C port or on the remote controller. The log can be exported through the computer and imported via the remote controller.

Power Button Module: Controls powering on or off of the battery station.

02 Material Identification — Battery Station



T30 Battery Charging Cable



Module



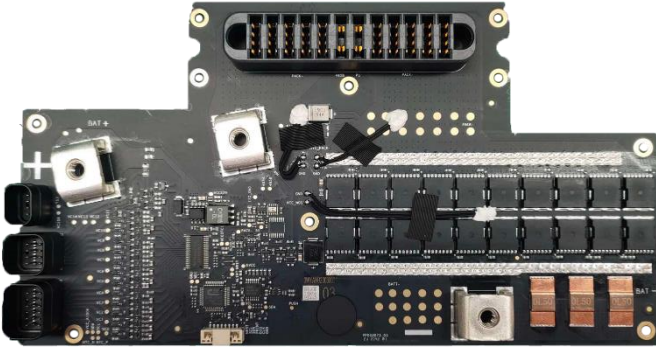
Power Module Adaptive Board

T30 Battery Charging Cable: An adaptive cable between the main board and battery, providing charging voltage and communication.

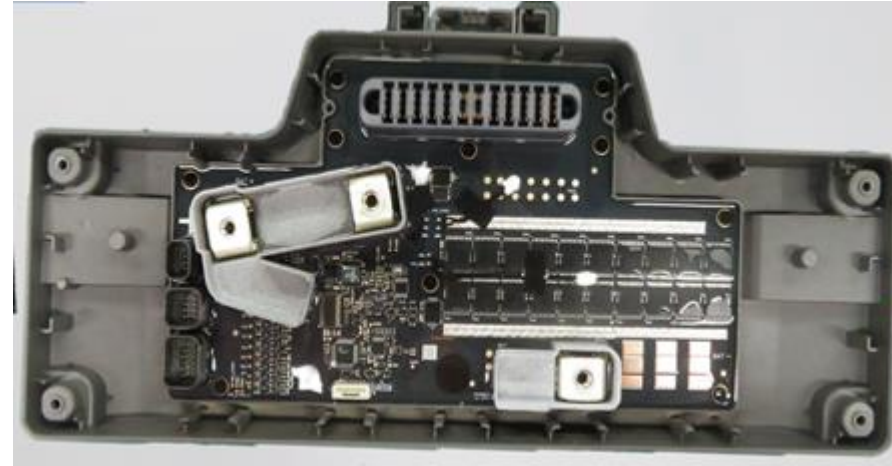
Power Module: The 220V electric power is converted into 59V direct current to charge the battery; also communicates with the main board. The maximum output current of a single power module can reach 65A.

Power Module Adaptive Board: Transfers power between the 4850 module and the main board. The power output of the power module is transferred to the main board through the 4850 module.

02 Material Identification — Battery



Battery Main Board



Battery Upper Shell Module

(Glue filling process to completely fix the battery main board in the upper shell)

Battery Main Board: Communicates with the battery station or aircraft, and manages and calculates the capacity of the battery cell module. The LED can display the battery status.

Upper Shell Module (Including battery adaptive board): Connects the battery cell, and connects the battery main board to the external charger and aircraft.



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03 Repair Level

03 Repair Strategy

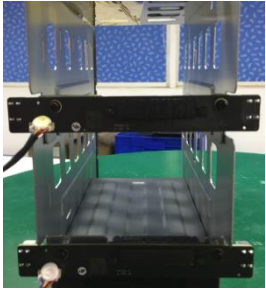
Product	Material	Dealers
Battery Station	Power Module	Replace the whole module.
	Main Board	Replace the whole module.
	LED Board	Replace the whole module.
	Power Module Adaptive Board	Replace the whole module.
	Structure Part	Repair all parts.
Battery	Battery Cell Module	Not related
	Upper Shell Module (Including Battery Main Board)	Not related
	Other Appearance Parts (Battery Level Sticker, Handle, Bottom Pad)	Replace the appearance part only, such as battery handle, battery level sticker, and bottom pad.



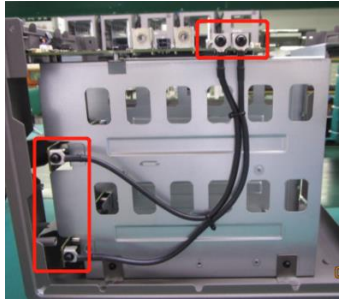
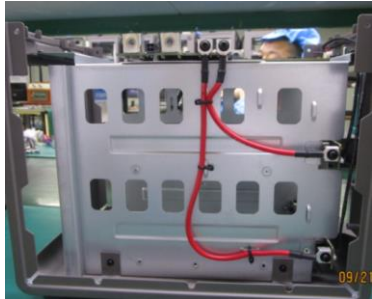
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04 Disassembly and Assembly Notices

04 Disassembly and Assembly Notices — Battery Station

No.	Item	Notice	Picture
1	White Glue	Apply white glue to the flat cable connector.	
2	Power Button	No fool-proof design; make sure the power icon faces upward.	
3	Waterproof Label	If there is no waterproof label on a received port board, attach a waterproof label if the port board needs to be replaced.	
4	Communication Cable	The communication cable needs to be attached first, and then mount the bracket module to the U-type frame.	

04 Disassembly and Assembly Notices

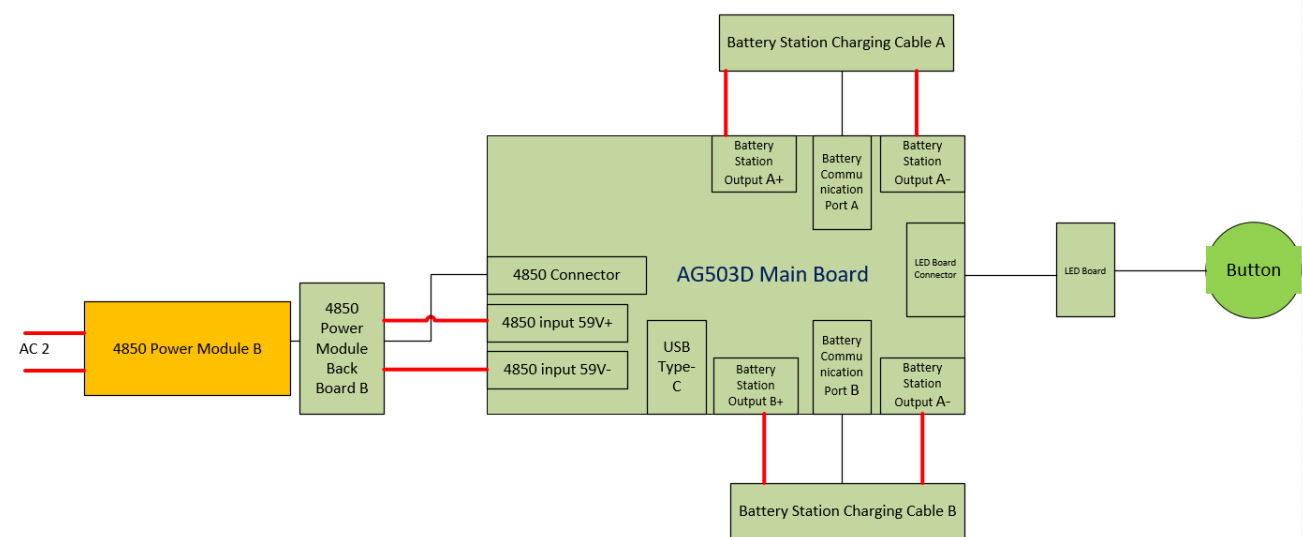
No.	Item	Notice	Picture
5	Torque	10.0±0.5kgf.m	
		8.0±0.5kgf.m	
		8.0±0.5kgf.m	



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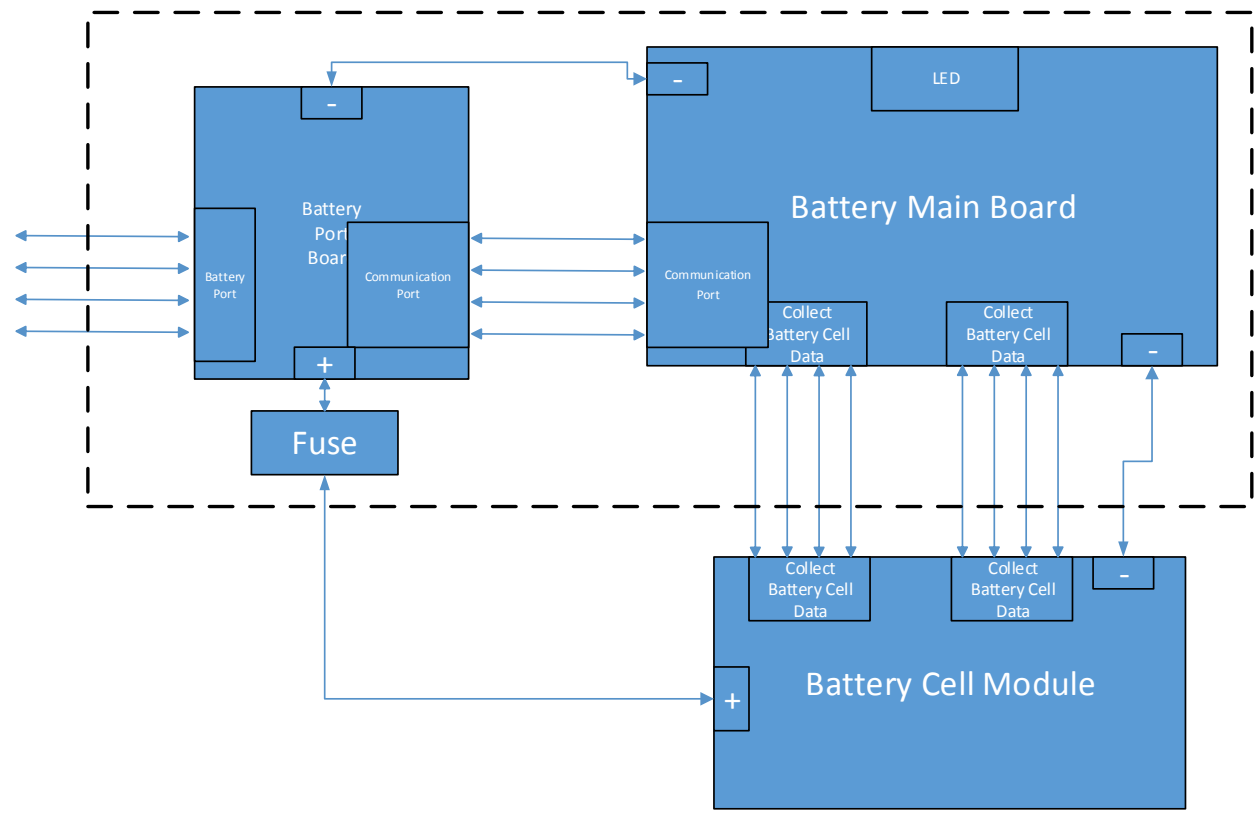
05 Principles Diagram and Boards Introduction

05 Principles Diagram – Battery Station



Chip	Function
4850 Power Module	The 220V electric power is converted into 59V direct current to charge the battery; also communicates with the main board. The maximum output current of a single power module can reach 65A.
4850 Module Power Back Board	Transfers power between the 4850 module and the main board. The power output of the power module is transferred to the main board through the 4850 module. The 4850 module communicate with the main board through the back board.
Battery Station Main Board	Communicates with the power module and battery, manages the battery charging current, and outputs the LED signal to control the LED display. The management board firmware can be updated by connecting to the computer via Type-C port or on the remote controller. The log can be exported through the computer and imported via the remote controller.
Output Cable	An adaptive cable between the main board and battery, providing charging voltage and communication.
LED Board	Connects the management board and the system switch respectively, and transmits the switch signal to the management board through the LED board. The LED on the LED board is used to display the battery charging status and the status of the battery station.

05 Principles Diagram – Battery



Chip	Function
Battery Cell Module	The core part of the battery and the source of battery power supply and charging. Main parameters: Number of cycles, internal resistance, voltage difference, which are used to judge the quality of the battery cell. Provided by two suppliers: ALT (number of cycles: 1000) and Desay (number of cycles: 1000)
Fuse	When the interface is short-circuited or abnormal, the current is too large and the fuse will be burnt out to protect the battery.
Battery Main Board	Communicates with the battery station or aircraft, and manages and calculates the capacity of the battery cell module. The LED can display the battery status.
Battery Adaptive Board	Connects the battery cell, and connects the battery main board to the external charger and aircraft.



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06 Test Plan & Fixtures

06 Test Plan & Fixtures – Battery Station

Type	Module	Test Item	Test Required After Component Replacement						
			Battery Station						
			Main Board	LED Board	Power Button	Positive and negative Cables	Adaptive Board	Power Module	Charging Cable
Battery Station	Whole Module	Fast and slow charging and full charge tests	√	√	√	√	√	√	√

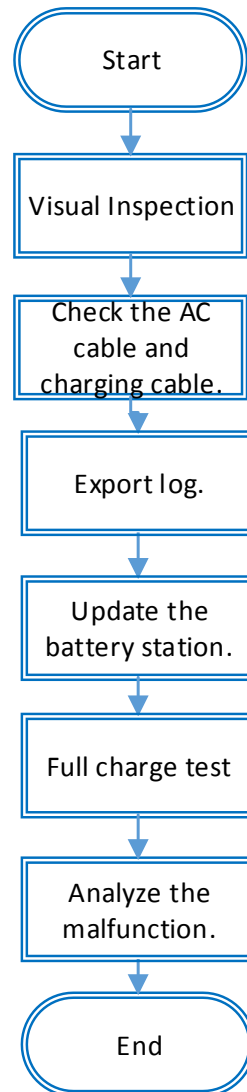


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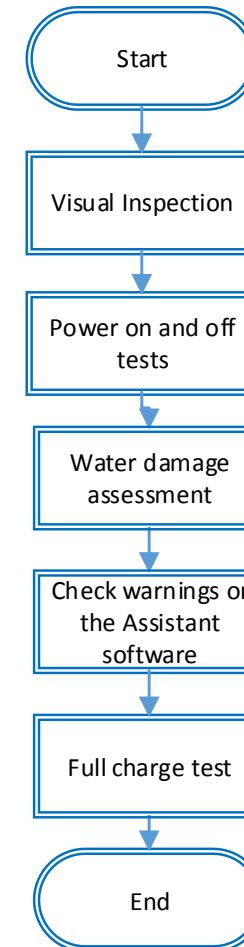
07 Malfunction Assessment and Analysis

07 Malfunction Assessment and Analysis — Battery Station and Battery

Battery Station Damage Assessment Process



Battery Damage Assessment Process



07 Malfunction Assessment and Analysis — Battery Station and Battery

The damage assessment process of the T30/T10 battery station is the same as that of the T16/T20 battery station.

During charging, the status LED will blink when the corresponding issue occurs.

LED Blinking Pattern	Warning Sound	Description	Component Replacement
Solid yellow	/	The main board is malfunctioning and the power is limited.	Main Board
Blinks yellow once	/	CPX-3000W power module is malfunctioning and the power limits.	Power Module
Blinks red once continuously	/	CPX-3000W power module cannot be used or the fan does not rotate.	Power Module or Fan
Blinks red once continuously	D.D.D...	CPX-3000W power module critical error detected, no output.	Power Module
Solid red	DD.DD.DD...	The main board is malfunctioning and cannot be used.	Main Board

Malfunction	Warning Sound	Description	Component Replacement
Power on and make alert	/	The product sounds an alert when powering on.	Main board or power module
Unable to power on	/	The product cannot be powered on.	Main board or power module
Make alert during charging	/	One channel or two channels	Main board or power module
Detect that the battery cannot be charged	/	The battery cannot be charged.	Main board or power module
The battery is charged but the battery level is not increased.	/	The battery level cannot increase.	Positive and negative or power module
The fan does not rotate.	/	The fan does not rotate.	Main board or power module
The LED does not light up.	/	The LED does not light up after powering on.	LED Board
The USB cable is malfunctioning.	/	The Assistant software or remote controller cannot be connected via the USB cable.	Main BOARD
	/	The product cannot be updated.	Main board
Power on and make noise	/	Normal inspection result or the product cannot be powered on.	Power Module
The cable is scorched.	/	The cable is discolored or scorched.	Positive and negative or power module



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FAQ

Compatibility List

Product	T30	T10	T20
T30 Intelligent Flight Battery	Supported	\	\
T10 Intelligent Flight Battery	\	Supported	\
T20 Intelligent Flight Battery	\	Supported	Supported
D9000i Multifunctional Inverter Generator	Supported	\	\
D4500i Multifunctional Inverter Generator	\	Supported	Supported
T30 Intelligent Battery Station (7200W)	Supported	\	\
T10 Intelligent Battery Station (3600W)	\	Supported	Supported
T20 4-Channel Intelligent Battery Charger	\	Supported	Supported



Thank You